AYURLEAF VISION: AI-BASED SYSTEM FOR AYURVEDIC PLANT DISEASE DETECTION AND ANALYSIS

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MOTIVATION / INTRODUCTION

- Traditional disease diagnosis of ayurvedic plants Neem, Tulsi, Turmeric, and Aloe Vera depends entirely on manual expert inspection, which is slow, error-prone, and unavailable in rural farming areas. No AI-based end-to-end diagnostic system exists specifically for ayurvedic species. AyurLeaf AI fills this gap with an intelligent, accessible, and explainable diagnostic pipeline.
- These plants are not just crops — they are the raw material for affordable healthcare used by millions who cannot access modern medicine. A diseased plant does not only mean financial loss for the farmer; it directly shrinks the availability of plant-based medicine for the communities that depend on it most.

SDG GOAL NUMBER:

SDG 3 — Good Health and Well-being (Primary) & SDG 15 — Life on Land (Secondary)

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OBJECTIVES

- Train and Classify **18 disease and healthy classes** across 4 ayurvedic species using MobileNetV2 transfer learning targeting accuracy above 85%.
- Localise and stage disease spread using **HSV-based lesion segmentation** into Low, Medium, and High severity levels with **Grad-CAM heatmaps** visually confirming model attention on actual disease lesion regions.
- Deliver the complete pipeline through **AyurLeaf AI**, a browser-based web application requiring zero technical expertise with environmental parameters — temperature, humidity, soil moisture — and disease severity staging.

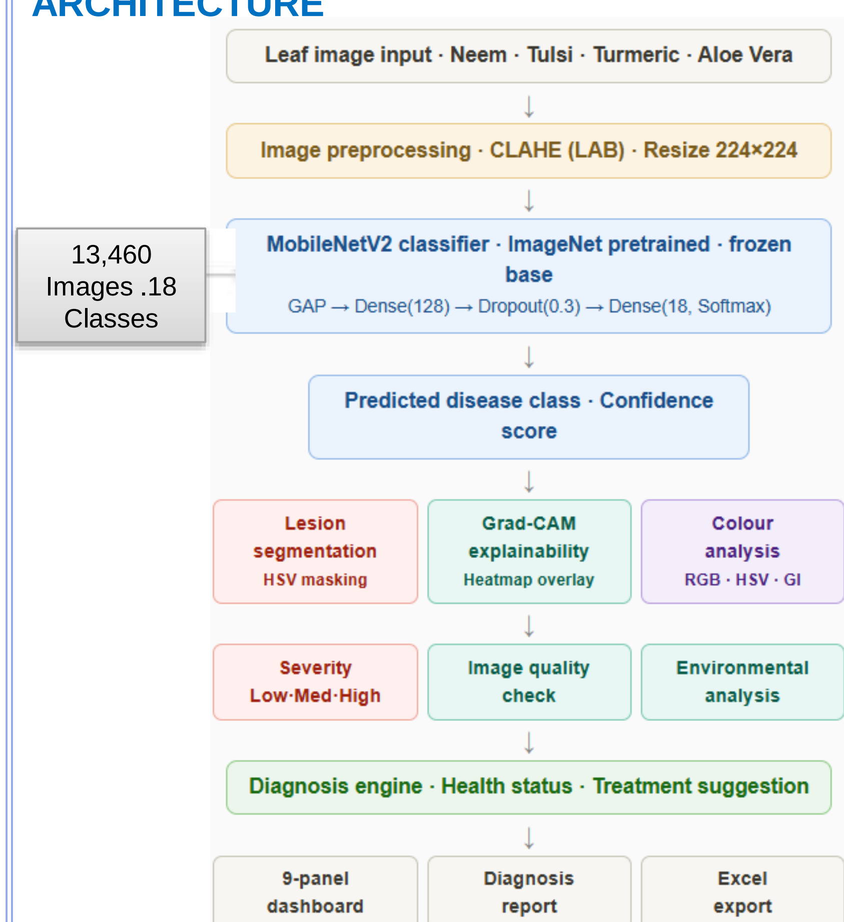
SCOPE OF THE PROJECT

- Dataset of 13,460 leaf images split 70/20/10 for training, validation, and testing.
- Full pipeline from raw image to structured diagnosis report with Excel export.
- Deployed as **AyurLeaf AI** — a browser-based web application covering Home, Analyze & Performance Dashboard requiring no installation and no ML expertise for field use.

METHODOLOGY

1. **Dataset Preparation** 13,460 leaf images across 18 classes split into 9,417 training, 2,686 validation, and 1,357 test images using stratified splitting.
2. **2. Image Preprocessing & Augmentation** CLAHE applied on LAB luminance channel to enhance disease-region contrast. Augmented with rotation, zoom, and horizontal flip.
3. MobileNetV2 pretrained on ImageNet with **frozen convolutional base** preserving learned visual features.
4. **Lesion Segmentation & Severity** HSV masking isolated brownish-yellow lesions. Severity ratio = lesion pixels / total pixels → Low (<10%), Medium (10–30%), High (>30%).
5. **Grad-CAM Explainability** Gradients from out_relu layer produced spatial heatmaps confirming model attention directly on disease lesion regions.
6. **Environmental & Colour Analysis** Temperature, humidity, and soil moisture interpreted through threshold rules. Greenness Index extracted per image to assess leaf health.
7. **Web Application — AyurLeaf AI** 4-page browser-based app — Home, Analyze, Performance Dashboard, Disease Library — with live diagnosis, probability charts, and Excel export.

ARCHITECTURE



RESULTS

CONCLUSION

AyurLeaf AI demonstrates that deep learning can be effectively applied to ayurvedic plant disease detection — achieving **90.13% test accuracy** across 18 classes, making it lightweight enough for real-world field deployment. The integration of CLAHE preprocessing, HSV severity staging, Grad-CAM explainability, and environmental analysis into a single unified pipeline goes beyond classification — delivering a complete, trustworthy, and actionable diagnostic system. Deployed as a browser-based web application, AyurLeaf AI makes intelligent plant disease diagnosis accessible to farmers and herbalists without any technical expertise.

CONTACT DETAILS

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- A. R. Revathi, N. B, S. M. Vinod and N. J. Warrier, "An Efficient Transfer Learning Approach for Identification and Severity Prediction of Plant Diseases," *2025 11th International Conference on Communication and Signal Processing (ICCSP)*, Melmaruvathur, India, 2025, pp. 572-577, doi: 10.1109/ICCSP64183.2025.11088680.